Geodata Viewer: supported data types

Which formats the viewer opens, how each can be reached, and how it is shown on the globe.

Opening routes: **On server**: a path on the machine running the viewer, read in place (Browse..., path box or ?url=). **Remote URL**: http(s):// or s3://, read in place. **From your computer** (viewer hosted elsewhere): **Browser** = read by your browser, never leaves your computer; **Temp upload** = sent to the server, deleted after 24 hours.

Key: Supported | Via temporary upload | Not supported

Type	Extensions	Recognised by	On server	Remote URL	From your computer	Shown as
GeoJSON	.geojson, .json	Extension, or content starting with {	Yes	Yes, up to 500 MB; also WFS / API links	Browser	Vector, reprojected to WGS84
Shapefile	.shp + .dbf / .shx / .prj, or .zip	Extension, or zip signature	Yes	Yes; sidecars fetched next to the .shp	Browser (pick the .shp with its sidecars)	Vector, reprojected using the .prj
GeoTIFF / COG	.tif, .tiff	Extension, or TIFF / BigTIFF signature	Yes, any GeoTIFF; COGs are fastest	COG via http(s) or s3	Browser (single image, max 3072 px); falls back to temp upload	Tiles: RGB as is; single band with colour map and min/max controls
PMTiles: raster	.pmtiles	Extension, or PMTiles signature	Yes	http(s), s3 (private buckets too)	Browser, tile by tile	Image tiles (PNG, JPEG, WebP, AVIF)
PMTiles: vector	.pmtiles	As above	Yes	Yes	Temp upload	Drawn to images on the server, one colour per layer
PMTiles: terrain	.pmtiles (quantized-mesh)	Metadata: quantized-mesh-1.0	Yes	Yes	Temp upload	Replaces the globe's terrain (one at a time)
NetCDF	.nc, .nc4, .netcdf, .cdf	Extension; /thredds/dodsC/ or /opendap/ in the URL; NetCDF / HDF5 signature	Yes	OPeNDAP, plain https (byte ranges), public s3	Temp upload	Tiles with variable picker, time / depth sliders, colour map
Zarr (v2 / v3)	.zarr folder	Folder containing .zgroup / .zarray / .zmetadata / zarr.json	Yes	https, public s3 (large stores can be slow)	Temp upload as a zipped .zarr	Same as NetCDF
Point cloud	.las, .laz, .copc.laz	Extension, or LASF signature	Yes; converted once, then cached	No; download it first	Temp upload (conversion takes minutes)	3D Tiles points (RGB if present)

Common to all types

- **CRS**: read from the file. Enter a source CRS only if the file has none or the wrong one. PMTiles are always Web Mercator; the browser's GeoTIFF reader needs an EPSG code.
- **The 180 degree line**: handled for every type.
- **Gridded data (NetCDF / Zarr)**: only regular 1-D lat/lon grids; curvilinear grids are rejected.
- **Forcing a type**: when a name or URL is ambiguous, add &type= with vector, raster, netcdf, zarr, pmtiles or pointcloud.